Fraud Detection | Machine Learning Case

End-to-end classification of fraudulent and legitimate transactions.

Historical development: 2025 | Scoring population: January 2026

Business goal: reduce undetected fraud while protecting legitimate customers.

01 | Data and validation

523,525 historical transactions; 20,485 labeled fraudulent (3.91%).

44,451 January 2026 transactions have no ground-truth fraud label.

Chronological split: 70% training, 15% validation, 15% out-of-time testing.

No direct customer identifiers used as predictors.

02 | Modeling approach

Compared logistic regression and histogram gradient boosting.

Validation PR-AUC: Gradient Boosting 0.668 vs Logistic Regression 0.255.

Selected Gradient Boosting and chose the decision threshold on validation only.

Threshold: 0.2828, selected to maximize validation F1.

03 | Out-of-time test performance

PR-AUC: 0.633 | ROC-AUC: 0.964

Precision: 57.4% | Recall: 65.6% | F1: 0.612

Confusion matrix: TN 73,053 | FP 1,791 | FN 1,268 | TP 2,417.

Interpretation: a material share of fraud is detected, but false alerts and missed fraud remain important.

04 | January 2026 fraud indicator

44,451 transactions scored; 2,350 (5.29%) flagged for investigation.

Risk tiers: High 2,350 | Medium 1,438 | Low 40,663.

High-risk transaction value: 70,369.44 (dataset currency units).

Columns delivered: fraud_probability, fraud_indicator, predicted_class, risk_level.

05 | Business recommendations

Prioritize high-risk transactions for analyst review, not automatic rejection.

Use medium-risk cases for secondary checks when review capacity permits.

Confirm the internal score is available before the fraud decision.

Monitor precision, recall, drift, customer friction and monetary costs.

06 | Conclusions and caveats

Gradient Boosting is the best tested model for this dataset under PR-AUC.

The fraud indicator identifies 5.29% of January activity as suspicious.

January predictions are NOT confirmed fraud events; January accuracy cannot be calculated without labels.

Next: validate operational costs, calibrate scores, monitor drift, and retrain with verified outcomes.